

Tide retrospective:
*Integrability for the generic- k monomial template (A1
follow-up)*

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Abstract

A1's retrospective deferred the integrability layer (`kth_integrable_pow*`) on the grounds that the Gaussian- domination argument needed case analysis on $k = 1$ vs $k \geq 2$. This Tide closes the layer with a uniform argument that avoids the case split: the key bound $x^2 \leq 1 + x^{2k}$ for $k \geq 1$ is immediate (case-split on $|x| \leq 1$ vs $|x| \geq 1$, the latter via `le_self_pow0`). Then refactors the quartic and sextic integrability lemmas as $k = 2$ / $k = 3$ specialisations. Net -17 lines across the three files; build clean (8298 jobs), zero `sorry` / `axiom` / `native_decide`.

1 Setting

A1 (generic even-monomial 1D template) landed nine theorems parametrised by $k \geq 1$: the moments, the partition function, and the expected values. The integrability witnesses¹ were deferred to a follow-up tide. A1's retrospective worried that the seabed's Gaussian- domination argument relied on k -specific square-completion identities ($k = 2$: $-6/t$; $k = 3$: $-\sqrt{240/t} \cdot 2/3$); a uniform formula at generic k would require a case-split or a parametric `Real.rpow` constant.

In fact a simpler approach exists, recorded here.

2 The thing formalised

A private bound:

- `sq_le_one_add_pow_two_mul` (private): for $k \geq 1$ and $x : \mathbb{R}$, $x^2 \leq 1 + x^{2k}$.

Two public theorems:

- `kth_integrable_pow`: for $k \geq 1$, $n : \mathbb{N}$, $t > 0$, the function $x^n \cdot \exp(-tx^{2k}/(2k)!)$ is Lebesgue integrable on $\mathbb{R}$.
- `kth_integrable_pow_pot`: same in `kthPotential` form.

Plus four refactors in the quartic and sextic files: the four `_integrable_pow*` theorems become thin specialisations.

¹Named `kth_integrable_pow` and `kth_integrable_pow_pot`.

3 Proof strategy

The bound $x^2 \leq 1 + x^{2k}$ for $k \geq 1$ uses an unconditional case split on $|x|$:

- $|x| \leq 1$: $x^2 \leq 1 \leq 1 + x^{2k}$ since $x^{2k} = (x^2)^k \geq 0$.
- $|x| \geq 1$: $x^{2k} = (x^2)^k \geq x^2$ by `le_self_pow0` applied to $x^2 \geq 1$ and $k \geq 1$. Then $1 + x^{2k} \geq x^2$.

From the bound, $tx^{2k}/(2k)! - (t/(2k)!)x^2 \geq -t/(2k)!$, hence $\exp(-tx^{2k}/(2k)!) \leq \exp(t/(2k)!) \cdot \exp(-(t/(2k)!)x^2)$. The dominator $|x|^n \cdot \exp(-(t/(2k)!)x^2)$ is Gaussian-times- poly, integrable by Mathlib’s `integrable_rpow_mul_exp_neg_mul_sq` with parameter $b = t/(2k)!$. `Integrable.mono` closes.

4 Roadblocks and resolutions

Three small fixes during the proof:

1. Missing import: the Mathlib Gaussian-integral file was needed for the dominator’s integrability.²
2. `le_self_pow` renamed to `le_self_pow0` in the current Mathlib snapshot.
3. `nlinarith` couldn’t close the bound directly because of the implicit identity $tx^{2k}/\text{fac} = (t/\text{fac}) \cdot x^{2k}$; added an explicit `field_simp`-based rewrite.

Each was a small local fix; none required deliberation.

5 What was Mathlib, what was new

From Mathlib. The Gaussian-times-poly dominator, `le_self_pow0` for the $(x^2)^k \geq x^2$ step, `Integrable.mono` and `.const_mul`, plus the standard `Real.exp` and absolute-value bookkeeping.

From the seabed. A1’s `kthPotential` definition. The quartic and sextic integrability lemmas use the new generic theorems as specialisations.

New here. The private bound $x^2 \leq 1 + x^{2k}$ and the two integrability theorems, plus the four specialisations in quartic / sextic.

6 Lessons

Case-split on k is not always necessary for k -parametric arguments. A1’s retrospective expected the integrability proof to require a case-split on $k = 1$ vs $k \geq 2$ because the seabed’s quartic and sextic proofs use k -specific square-completion identities. In fact the simpler bound $x^2 \leq 1 + x^{2k}$ works uniformly — it just needed to be found. The lesson: when a follow- up tide looks blocked by a case-split, search for a coarser uniform bound that avoids the case-split entirely.

²Specifically `Mathlib.Analysis.SpecialFunctions.Gaussian.GaussianIntegral` for `integrable_rpow_mul_exp_neg_mul_sq`.

`nlinarith` sees **Real.rpow-style arithmetic better than Real.rpow-styled goals**. The first build attempt had `nlinarith` fail to close a step that turned on the identity $tx^{2k}/fac = (t/fac) \cdot x^{2k}$. An explicit `field_simp`-rewrite to put the goal in the latter form made the rest of `nlinarith` routine. Worth recording.

Mathlib renaming. `le_self_pow` is now `le_self_pow_0` in the current Mathlib snapshot. Other `_0`-suffixed renames exist (`pow_le_pow_right_0`, etc.) for the `GroupWithZero`-class versions. Use the `_0` variants when working with $\mathbb{R}$ -valued bounds.

7 Follow-ups

- *Optional: tighten the bound.* The bound $x^2 \leq 1 + x^{2k}$ is coarse; for $k \geq 2$, tighter bounds with k -dependent constants are available (e.g. the $((k-1)/k) \cdot ((2k)!/(kt))^{1/(k-1)}$ minimum). For downstream consumers that need a tight asymptotic constant, the present bound is enough; for purely integrability use, the coarse bound suffices.
- *Promote to `lean-common`.* If a third consumer of the bound $x^2 \leq 1 + x^{2k}$ materialises outside the laplace repo, factor it up. Currently one use site.
- *Multi-index analogues.* 2D / multi-index generalisations of the integrability bound would follow the same template; deferred until a downstream consumer asks.

Acknowledgements

No GPT consult was used (A1-integrability is a focused follow-up of A1, and A1's pre-consult had set the relevant scope decisions). The uniform Gaussian-domination bound was found by inspection of the seabed's sextic argument and noting that $x^2 \leq 1 + x^6 \Rightarrow x^2 \leq 1 + x^{2k}$ for any $k \geq 1$.